

Seminar Nr. 3

FUNȚIILE LUI EULER Γ ȘI β

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Probleme rezolvate

1. Să se calculeze următoarele integrale, folosind funcția Γ :

i) $\int_0^\infty x^3 e^{-x} dx$, ii) $\int_0^\infty \sqrt{x} e^{-x} dx$, iii) $\int_0^\infty \frac{e^{-x}}{\sqrt{x}} dx$,

iv) $\int_1^\infty x \sqrt[3]{x^2 - 1} e^{-x^2} dx$, v) $\int_{-\infty}^0 x^4 e^x dx$.

Soluție. i) Integrala $\int_0^\infty x^3 e^{-x} dx$ se poate calcula folosind funcția Γ :

$$\int_0^\infty x^3 e^{-x} dx = \Gamma(4) = 3! = 6.$$

ii) Integrala $\int_0^\infty \sqrt{x} e^{-x} dx$ nu mai poate fi calculată cu tehnicile învățate la liceu, deci vom folosi funcția Γ a lui Euler.

$$\int_0^\infty \sqrt{x} e^{-x} dx = \int_0^\infty x^{\frac{1}{2}} e^{-x} dx = \Gamma\left(\frac{1}{2} + 1\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

iii) Se observă că

$$\int_0^\infty \frac{e^{-x}}{\sqrt{x}} dx = \int_0^\infty x^{-\frac{1}{2}} e^{-x} dx = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

iv) Se face schimbarea de variabilă $x^2 - 1 = t$. Atunci $t \in [0, \infty]$, iar $2x dx = dt$. Cu acestea integrala devine

$$\int_1^\infty x \sqrt[3]{x^2 - 1} e^{-x^2} dx = \frac{1}{2} \int_0^\infty \sqrt[3]{t} e^{-t-1} dt = \frac{1}{2e} \Gamma\left(\frac{1}{3} + 1\right) = \frac{1}{6e} \Gamma\left(\frac{1}{3}\right).$$

v) Prin simpla schimbare $x \mapsto -x$, avem

$$\int_{-\infty}^0 x^4 e^x dx = \int_0^\infty x^4 e^{-x} dx = \Gamma(5) = 4! = 24.$$

2. Folosind proprietățile funcțiilor Γ și β , să se calculeze valoarea următoarelor integrale:

i) $\int_0^1 x^2 \sqrt[3]{1-x} dx$; ii) $\int_0^1 x^3 \sqrt{1-\sqrt{x}} dx$; iii) $\int_0^1 x^{10} (1-x)^{15} dx$;
 iv) $\int_0^{\frac{\pi}{2}} \sqrt[3]{\sin x} dx$, v) $\int_0^\infty \frac{\sqrt[3]{x}}{1+x^2} dx$; vi) $\int_0^{\frac{\pi}{2}} \frac{\sin^5 x}{\sqrt[6]{\cos x}} dx$.

Soluție. i) Integrală se poate calcula cu ajutorul funcțiile speciale ale lui Euler Γ și β .

$$\begin{aligned} \int_0^1 x^2 \sqrt[3]{1-x} dx &= \beta\left(3, \frac{4}{3}\right) = \frac{\Gamma(3)\Gamma\left(\frac{4}{3}\right)}{\Gamma\left(3+\frac{4}{3}\right)} = \frac{2!\Gamma\left(1+\frac{1}{3}\right)}{\Gamma\left(1+\frac{10}{3}\right)} = \frac{2 \cdot \frac{1}{3}\Gamma\left(\frac{4}{3}\right)}{\frac{10}{3}\Gamma\left(\frac{10}{3}\right)} \\ &= \frac{1}{5} \frac{\Gamma\left(\frac{1}{3}\right)}{\Gamma\left(1+\frac{7}{3}\right)} = \frac{\Gamma\left(\frac{1}{3}\right)}{\frac{5 \cdot 7}{3} \cdot \Gamma\left(\frac{7}{3}\right)} = \frac{\Gamma\left(\frac{1}{3}\right)}{\frac{5 \cdot 7 \cdot 4}{3^2} \cdot \Gamma\left(\frac{4}{3}\right)} = \frac{\Gamma\left(\frac{1}{3}\right)}{\frac{5 \cdot 7 \cdot 4}{3^3} \cdot \Gamma\left(\frac{1}{3}\right)} = \frac{27}{140}. \end{aligned}$$

ii) Substituția $x = t^2$ conduce la $dx = 2t dt$ și exprimă integrala cu ajutorul funcției β .

$$\begin{aligned} \int_0^1 x^3 \sqrt{1-\sqrt{x}} dx &= \int_0^1 2t^7 (1-t)^{\frac{1}{2}} dt = 2\beta\left(8, \frac{3}{2}\right) \\ &= 2 \frac{\Gamma(8)\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(8+\frac{3}{2}\right)} = 2 \frac{7! \frac{1}{2}\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(9+\frac{1}{2}\right)} = \frac{2^9 \cdot 7!}{17!!}. \end{aligned}$$

$$\text{iii)} \int_0^1 x^{10} (1-x)^{15} dx = \beta(11, 16) = \frac{\Gamma(11)\Gamma(16)}{\Gamma(27)} = \frac{10! \cdot 15!}{26!}.$$

iv) Cu substituția $\sin x = \sqrt{t}$, se obține

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sqrt[3]{\sin x} dx &= \int_0^1 \sqrt[3]{\sqrt{t}} \cdot \frac{1}{\sqrt{1-t}} \cdot \frac{1}{2\sqrt{t}} dt = \frac{1}{2} \int_0^1 t^{-\frac{1}{3}} (1-t)^{-\frac{1}{2}} dt \\ &= \frac{1}{2} \beta\left(\frac{2}{3}, \frac{1}{2}\right) = \frac{1}{2} \frac{\Gamma\left(\frac{2}{3}\right) \cdot \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{7}{6}\right)} = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{2}{3}\right)}{\Gamma\left(\frac{7}{6}\right)} = 3\sqrt{\pi} \frac{\Gamma\left(\frac{2}{3}\right)}{\Gamma\left(\frac{1}{6}\right)}. \end{aligned}$$

v) Schimbarea de variabilă $x^2 = \frac{t}{1-t}$ conduce la $x = \sqrt{\frac{t}{1-t}}$ de unde rezultă $dx = \frac{1}{2} \cdot \frac{\sqrt{1-t}}{\sqrt{t}} \cdot \frac{1}{(1-t)^2} dt$. Pentru $x = 0 \Rightarrow t = 0$, iar pentru $x = \infty \Rightarrow t = 1$. Prin urmare, integrala devine

$$\begin{aligned} \int_0^\infty \frac{\sqrt[3]{x}}{1+x^2} dx &= \frac{1}{2} \int_0^1 t^{-\frac{1}{3}} (1-t)^{-\frac{2}{3}} dt = \frac{1}{2} \beta\left(\frac{2}{3}; \frac{1}{3}\right) = \\ &= \frac{1}{2} \frac{\Gamma\left(\frac{2}{3}\right) \Gamma\left(\frac{1}{3}\right)}{\Gamma(1)} = \frac{1}{2} \Gamma\left(\frac{1}{3}\right) \Gamma\left(1 - \frac{1}{3}\right) = \frac{1}{2} \cdot \frac{\pi}{\sin \frac{\pi}{2}} = \frac{\pi\sqrt{3}}{3}. \end{aligned}$$

vi) Cu schimbarea de variabilă $x = \arcsin \sqrt{t}$, se obține

$$\int_0^{\frac{\pi}{2}} \frac{\sin^5 x}{\sqrt{\cos x}} dx = \frac{1}{2} \beta\left(3; \frac{5}{12}\right) = \frac{1}{2} \cdot \frac{\Gamma(3) \Gamma\left(\frac{5}{12}\right)}{\Gamma\left(\frac{41}{12}\right)} = \frac{1728}{2465},$$

deoarece

$$\Gamma\left(\frac{41}{12}\right) = \Gamma\left(1 + \frac{29}{12}\right) = \frac{29}{12} \Gamma\left(1 + \frac{17}{12}\right) = \frac{29}{12} \cdot \frac{17}{12} \Gamma\left(1 + \frac{5}{12}\right) = \frac{29 \cdot 17 \cdot 5}{12^3} \Gamma\left(\frac{5}{12}\right).$$

Probleme propuse

1. Folosind funcțiile lui Euler, să se calculeze următoarele integrale:

i) $I_1 = \int_0^\infty x^8 e^{-2x} dx$; $I_2 = \int_0^\infty x^{18} e^{-x^2} dx$; $I_3 = \int_0^\infty x^{28} e^{-x^3} dx$;

ii) $I_4 = \int_{-\infty}^{-5} (x+5)^8 e^{x+5} dx$; $I_5 = \int_{-\infty}^2 (x-2)^{17} e^{x-3} dx$;

$I_6 = \int_{-\infty}^{-1} (x+1)^{23} e^{x+23} dx$; $I_7 = \int_0^1 x (\ln x)^7 dx$;

iii) $I_8 = \int_0^\infty \sqrt{x^{16} - x^{17}} dx$; $I_9 = \int_0^\infty x^{23} \sqrt{1-x} dx$;

$$\begin{aligned}
I_{10} &= \int_{-1}^0 x^{43}(1+x)^{34}dx; \\
\text{iv)} \quad I_{11} &= \int_0^7 x^2 \sqrt{49-x^2}dx; \quad I_{12} = \int_0^3 x^2 \sqrt{(9-x^2)^{11}}dx; \\
\text{v)} \quad I_{13} &= \int_0^{\frac{\pi}{2}} \sin^{43} x \cos^{34} x dx; \quad I_{14} = \int_0^{\frac{\pi}{2}} \frac{\sin^4 x}{\sqrt[3]{\cos x}} dx; \\
I_{15} &= \int_0^{\frac{\pi}{2}} \frac{\cos^{10} x}{\sqrt[8]{\sin x}} dx; \quad I_{16} = \int_0^{\frac{\pi}{2}} \cos^{54} x dx; \\
\text{vi)} \quad I_{17} &= \int_0^\infty \frac{x}{(1+x^3)^2} dx; \quad I_{18} = \int_0^\infty \frac{\sqrt{x}}{1+x^6} dx.
\end{aligned}$$

Bibliografie. D. Păunescu, A. Juratoni, *Calcul integral avansat*, Editura Orizonturi Universitare, Timișoara, 2015.